INCH-POUND

MIL-DTL-29431A(MC)
16 August 2023
SUPERSEDING
MIL-C-29431(MC)
4 June 1987

DETAIL SPECIFICATION

CAP, GARRISON, WOMEN'S, USMC

This specification is approved for use by the United States Marine Corps and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers the requirements for a women's garrison cap worn by U.S. Marine Corps personnel.

1.2 Classification. The cap is available in the following types, classes, and sizes, as specified (see 6.2).

1.2.1 Types.

Type I – Lined

Type II – Unlined (optional program)

1.2.2 Classes (see 6.5).

Class 1 – Polyester/wool gabardine, Green shade 2212

Class 2 – deleted

1.2.3 Sizes.

21 21-1/2 22 22-1/2 23 23-1/2 24

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

COMMERCIAL ITEM DESCRIPTIONS

A-A-50199 - Thread, Polyester Core, Cotton or Polyester-Covered

A-A-52106 - Cloth, Twill or Plain Weave, Polyester and Polyester Blend

Comments, suggestions, or questions on this document should be addressed to Marine Corps Systems Command, 2200 Lester Street, Quantico, VA 22134 ATTN: SEAL-SE-STDS or emailed to USMC_STDZ@usmc.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 8410

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DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-10176 - Cloth, Gabardine: Wool, Polyester and Wool

MIL-DTL-10296 - Cloth, Sateen, Cotton

MIL-DTL-32075 - Label: For Clothing, Equipage, and Tentage, (General Use)

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-902 - Provisions for Evaluating Quality of Caps, Garrison, Women's

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

CODE OF FEDERAL REGULATIONS

40 CFR 798 - Health Effects Testing Guidelines

(Copies of this document are available online at https://ecfr.gov.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AEROSPACE INDUSTRIES ASSOCIATION (AIA)

NASM20652 - Eyelets, Metallic, and Eyelet Washers, Metallic

NASM20652/1 - Eyelets, Metallic, Rolled Flange Type, and Eyelet Washer

(Copies of this document are available online at www.aia-aerospace.org.)

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC EP1 - Gray Scale for Color Change

AATCC EP2 - Gray Scale for Staining

AATCC TM8 - Colorfastness to Crocking: Crockmeter Method

AATCC TM15 - Colorfastness to Perspiration

AATCC TM20A - Test Method for Fiber Analysis: Quantitative

AATCC TM132 - Colorfastness to Drycleaning

AATCC TM158 - Dimensional Changes on Drycleaning, Perchloroethylene

(Copies of these documents are available online at www.aatcc.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at www.asq.org.)

ASTM INTERNATIONAL

ASTM D1907/D1907M - Standard Test Method for Linear Density of Yarn (Yarn Number) by the Skein Method

ASTM D2724 - Standard Test Method for Bond Strength of Bonded, Fused, and Laminated Apparel Fabrics

ASTM D3775 - Standard Test Method for End (Warp) and Pick (Filling) Count of Woven Fabrics

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ASTM D3776/D3776M - Standard Test Methods for Mass per Unit Area (Weight) of Fabric

ASTM D5034 - Standard Test Method for Breaking Strength and Elongation of Textile Fabrics (Grab Test)

ASTM D6193 - Standard Practice for Stitches and Seams

(Copies of these documents are available online at www.astm.org.)

ORGANISATION FOR ECONOMIC CO-OPERATION AND DEVELOPMENT (OECD)

OECD Test No. 404 - Acute Dermal Irritation/Corrosion

OECD Test No. 406 - Skin Sensitisation

(Copies of these documents are available online at www.oecd-ilibrary.org.)

SAE INTERNATIONAL

AMS-STD-595 - Colors Used in Government Procurement

(Copies of this document are available online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Inspections.

3.1.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.1.2 Conformance inspection. When specified (see 6.2), a sample shall be subjected to a conformance inspection in accordance with 4.3.

3.2 Design. The cap is in the U.S. Marine Corps design shown on figures 1 and 2.

3.2.1 Figures. Figures 1 through 3 are provided for information purposes only and are applicable to type I and type II. When inconsistencies exist between the written specification and the figures, the written specification shall govern.

3.3 Materials and components.

3.3.1 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3.2 Basic material. The basic material for the class 1 cap shall be a polyester/wool blend gabardine cloth, 6.8 ounces per square yard, Green shade 2212 (AMS-STD-595 chip number 34080) conforming to MIL-DTL-10176, type II, class 12, when tested as specified in 4.4.1.

3.3.3 Laminated material (for type II only). For the type II cap, the material for the crown, side crown, and curtain of the garrison cap shall be a laminate material consisting of basic material (see 3.3.2), a polyester polyurethane foam (see 3.3.3.1), and a polyester knit backing material (see 3.3.3.2). Under no circumstances shall the dry heat temperature of the lamination process exceed 315 °F. The bonded basic material/foam/knit material shall be in accordance with the requirements specified in table I when tested as specified in 4.4.1.

3.3.3.1 Foam. The foam for the basic material laminate shall be polyester polyurethane foam meeting the applicable requirements cited in table I. The color of the foam shall be charcoal. The backing and foam shall meet the toxicity requirements specified in 3.9 when tested as specified in 4.4.2.

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3.3.3.2 Knit. The knit backing for the basic material laminate shall be a 70 denier polyester interlock knit. The knit material shall conform to the requirements specified in table I, when tested as specified in 4.5.1, and the shade shall be a good match to the basic material (see 3.3.2). The knit shall meet the toxicity requirements in 3.9 and shall be tested as specified in 4.4.2.

TABLE I. Laminate material physical requirements.

Characteristic	Requirement
Basic material – polyester/wool gabardine	3.3.2
Foam	
Weight, ounces per square yard	0.07 to 1.20
Melt point,℃	290±5
Knit	
Weight, ounces per square yard	3.3±0.2
Yarns per inch:	
Wales	41±1
Courses	38±1
Colorfastness to perspiration (acid and alkaline)	3
Knit type	Interlock
Dimensional stability, wales and courses, percent (maximum) 1/	2

FOOTNOTE:

1/ There shall be no degradation, bubbling, puckering, or delamination after wet dry cleaning or steam pressing.

3.3.4 Lining (type I cap only).

3.3.4.1 Rayon lining. Rayon lining shall not be used on this item. Lining material meeting the requirements of 3.3.4.2 or 3.3.4.3 shall be used in lieu of rayon.

3.3.4.2 Polyester lining. A 100 percent polyester cloth conforming to A-A-52106, in a color that is a good match to Green shade 2209 (AMS-STD-595 chip number 34080), when tested as specified in 4.4.1, may be used.

3.3.4.3 Acetate lining. As an alternate to polyester cloth, a 100 percent acetate cloth conforming to the requirements of table II, when tested as specified in 4.4.1, may be used.

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TABLE II. Acetate lining (type I cap only) requirements.

Characteristic	Requirement
Fiber content	100 percent acetate
Denier (minimum)	150
Weave	5 harness satin weave and a counter of 2
Shade	A good match to AMS-STD-595 chip number 34080
Weight, ounces per square yard (minimum)	4.5
Yarns per inch (minimum)	
Warp	200
Filling	62
Breaking strength, pounds (minimum)	
Warp	100
Filling	50
Dimensional stability, percent (maximum)	±3
Colorfastness to, rating (minimum):	
Perspiration	4
Drycleaning	4
Crocking	
Wet	3.5
Dry	3.5

3.3.5 Sweatband. The material for the sweatband shall be a cotton sateen cloth conforming to MIL-DTL-10296, class 1, in a shade that is a good match to Olive Green shade 107, when tested as specified in 4.4.1.

3.3.5.1 Alternate sweatbands for type II cap only.

3.3.5.1.1 Sweatband, artificial leather. As an alternate for the type II cap, an artificial leather sweatband may be used, with approval of the material by the contracting agency (see 6.7).

3.3.5.1.2 Sweatband, cloth, interfaced. As an additional alternate for the type II cap, a cloth sweatband interfaced with a commercially acceptable woven interfacing may be used. The material for a cloth sweatband shall conform to MIL-DTL-10296, class 1, Olive Green 107, when tested as specified in 4.4.1. In the finished cloth sweatband, the upper edge of the sweatband shall be folded over the upper edge of the interfacing 1/4 to 3/8 inch and stitched in place. The finished cloth sweatband shall be 1-1/4±1/8 inches wide.

3.3.6 Thread. The threads shall conform to the requirements of the applicable thread commercial item descriptions cited in table III, when tested as specified in 4.4.1. Threads visible on the outside of the cap when worn shall be a good match to the basic fabric. Threads used to attach the sweatband that are not visible on the outside of the cap when worn shall be a good match to the basic material.

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TABLE III. Thread requirements.

Use (Seam Type)	Description	Document	Tex	Ply
Seaming (101, 301)	Cotton-covered or polyester-covered polyester core	A-A-50199, type I or II	50 (36-45) for needle and bobbin	2
Seaming (401)	Cotton-covered or polyester-covered polyester core	A-A-50199, type I or II	50 (36-45) for needle	2
			70 (31-35) for looper	2
Overedging (502, 503, 505)	Cotton-covered or polyester-covered polyester core	A-A-50199, type I or II	70 (31-35) for needle and looper	2
Raise, edge, and topstitching (101, 301)	Cotton-covered or polyester-covered polyester core	A-A-50199, type I or II	50 (36-45) for needle and bobbin	2

3.3.6.1 Colorfastness requirements. The threads shall conform to the colorfastness to light and drycleaning requirements specified in the applicable specification, when tested as specified in 4.4.1.

3.3.7 Eyelet and optional washer. The eyelet shall conform to the requirements of NASM20652 and NASM20652/1 with part number M20652/1-BBE108, except the color shall be enameled to approximate the basic material color. The optional washer shall conform to M20652/1-BBW101 in any color.

3.3.8 Labels.

3.3.8.1 Combination label. Each cap shall have a combination identification, instruction, and size label conforming to type VI, class 4 of MIL-DTL-32075, when tested as specified in 4.4.1, including the colorfastness to drycleaning requirement. The inscription shall be as indicated below. The label shall measure not less than 1-3/4 by 1-1/4 inches, with a minimum 1/4-inch margin at the top edge for sewing.

Nomenclature (item identification)

Contract number

Name of contractor

Size

National stock number (NSN)

Wool Product Act information

DRY CLEAN ONLY

3.3.8.2 Barcode label/tag. Each item shall be individually barcoded with a paper label/tag conforming to MIL-DTL-32075, type VIII, class 17 when tested as specified in 4.4.1. The tag shall be attached to the item by a fastener. The barcoding element shall be a 13-digit NSN and the appropriate 12-digit Universal Product Code (UPC) assigned for all NSNs by the government. The initials “UPC” shall appear beneath the code. The barcodes for NSN and UPC shall be a medium to high density and shall be located so that they are completely visible on the item when it is folded and packaged as specified. The tag’s location shall cause no damage or staining to the cap.

3.4 Patterns. Standard patterns, to be used to cut working patterns, will be furnished by the Government (see 6.2). The patterns are marked or notched for the proper cutting, placement, and assembly of the component parts. The working patterns shall be identical to the Government patterns. The Government patterns shall not be altered in any way. Minor modifications may be made to the working patterns where necessary, such as additional notches for use during construction when using automatic equipment. These modifications shall not alter the dimensional, serviceability, or appearance requirements cited in the document or those governed by the pattern. The standard patterns have a 1/4-inch seam allowance.

3.4.1 Pattern parts. The component parts of the cap shall be cut from the materials specified and shall be in accordance with the number of cut parts as specified in table IV.

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TABLE IV. Pattern parts.

Material	Nomenclature [1/]	Cut Parts [1/]	
		Type I	Type II
Basic material	CURTAIN	2	1
	SIDE CROWN	2	2
	TOP CROWN	1	1
Lining	CURTAIN LINING	1	-
	SIDE CROWN LINING	2	-
	TOP CROWN LINING	1	-

FOOTNOTE:

1/ The specific pattern parts are applicable to only the type of cap to be produced. Vendors shall verify that the proper part is used.

3.5 Shade and size marking. The component parts of the cap shall be marked or ticketed to ensure a uniform shade and size throughout the garment. Provided the numbering does not show on the outside of the garment; and wherever possible, is covered by the seam allowance, any method may be used except those listed below:

a. Corrosive metal fastening devices.

b. Adhesive type tickets, which show discoloration or leave traces of paper or adhesive on the material upon removal or adhere to the material upon removal.

c. Sewn-on shade tickets or metal fastening devices on the lining material.

3.6 Construction. Construction of the cap shall conform to the requirements specified in table V and table VI.

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TABLE V. Construction requirements.

Construction Feature	Requirement
Stitch, seam, and stitch type	In accordance with ASTM D6193 and table VI 1/
Stitches per inch	Minimum and maximum in accordance with table VI
Thread type and size	In accordance with table III
Thread tension	Maintained so that there is no tight, loose, or unbalanced stitching
Loops of 401 or 101 stitch type	Loops of the chain to finish on the inside of the cap
Overedging	Knife set to trim the loose threads and raveled edge without cutting the material
Seams and seam allowances	Maintained with seam uniformly sewn so that no raw edges, runoffs, twists, pleats, puckers, gathers, other distortions, or open seams result
Beginning and end of seams	Start and finish evenly
Ends of seams and stitching – 301 stitch type	If not caught in another line of stitching: Backstitched not less than 1/4 inch If the end of a continuous line of stitching: Overlapped not less than 1/2 inch.
Thread breaks (all stitch types)	Secured by stitching back of the break not less than 1/2 inch If 401 stitch type, repair using 301 stitch type
Skipped stitches (301 or 401)	Repaired by using 301 stitch type
Edge, raise, or topstitching	Unless otherwise specified in table VI, 1/16 to 1/8 inch from the edge
Additional basting	Allowed to facilitate manufacturing provided the basting thread does not show on the outside of the finished cap

FOOTNOTE:

1/ Whenever two or more methods for seam or stitch types are given for the same part of the operation, any one of the specified methods may be used.

3.6.1 Manufacturing operations requirements. The caps shall be made in accordance with all operation requirements specified in table VI. The contractor is not required to follow the exact sequence of operations as specified, provided the finished cap is identical to that produced by following the sequence of operations as specified in table VI. Minor modifications are permitted where necessary when using automatic equipment. Unless otherwise specified in table VI, operations are applicable to both types and seam allowances shall be as specified in 3.4.

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TABLE VI. Table of operations.

No.	Operation	Stitch Type	Seam Type	Stitches Per Inch
1	Cutting.			
	a. Cut the cap in strict accordance with the patterns furnished by the Government.			
	b. Cut the material in the direction indicated by directional lines on the patterns.			
	c. Cut the outershell of the cap from one piece of basic material.			
	d. Cut the lining for the type I cap from one piece of lining material.			
	e. Cut the cloth sweatband on the straight (warp or filling direction) from the specified material. The sweatband shall be cut 1-5/8 inches wide and sufficiently long in order to lap the ends 3/4 inch, with the top end turned under not less than 3/8 inch (see operation 12).			
	f. For the type II cap only: Cut the alternate sweatband to the required length (see operation 12).			
2	Replacement of damaged or defective parts.			
	a. At the time of cutting, replace the following: all parts of the basic material or lining material containing holes; any seriously weakening defects such as smashes, multiple floats, loose or weakening slubs expected to develop into a hole; and any shade bars, dye streaks or unsightly slubs, etc., affecting appearance.			
	b. During the manufacturing process, replace the following: any part damaged by needle chews that may be expected to develop into a hole; any scissors or knife cuts; any tear, hole, mend, burn, or large drill hole in the material affecting appearance.			
3	Shade marking.			
	a. Mark all component parts as specified in 3.5 in order to ensure a uniform shade and size throughout the cap and proper assembly.			
4	Make curtain.			
	a. Finished appearance. The curtain shall be uniformly stitched and shall be smooth and flat without twists, gathers, puckers, or pleats. The edge shall not be stretched. The edges shall be neatly and completely worked out.			
	b. The type 1 curtain shall be made of two plies of basic material. The type 2 curtain shall be made of one layer of laminated material with the knit backing facing the inside of the cap. All raw edges of type II cap curtain pieces shall be overedge stitched.	301 or 101	SSe-2(a)	12-14
	c. For type 1 curtain: Position the two plies face to face, with edges even and notches matching. Stitch together with a 1/4- to 5/16-inch seam along the upper edges.	301 or 101	Sse-2(b)	12-14

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TABLE VI. Table of operations – Continued.

No.	Operation	Stitch Type	Seam Type	Stitches Per Inch
4 (Cont'd)	d. Turn curtain, forcing out the seam. Uniformly edgestitch 1/8 inch from seamed edge.	301 or 101	Ssa-1	12-14
	e. Close bottom of curtain uniformly with a 3/16- to 1/4-inch seam.			
5A	Join top and side crowns (type I).			
	a. Finished appearance. The top and side crowns shall be joined together without twists, puckers, pleats, or distortion.			
	b. Position the left- and right-side crown on the outside of the respective top crown, with top edges even and the point at each end of the top crown matching the notches in the side crowns.			
	c. Join the left- and right-side crowns, respectively, to the top crown with a 1/4-inch seam, beginning and ending on the top crown 1/4 inch from each end.	301 or 101	LSq-2(a)	10-14 or 6-10
5B	Join top crown (type II) (see operation 13 for alternate).			
	a. Finished appearance. The top crown shall be joined together without twists, puckers, pleats, or distortion.			
	b. Fold top crown in half lengthwise with overedged edges even and notches matching. Stitch along folded edge from end; the line of stitching shall taper from zero at each end to 1/8 to 3/16 inch from folded edge at center.	301	Osf-1	8-12
	c. Stitch top crown 3/4±1/8 inch from overedged edge.	301	Ssa-1	8-12
6	Join side crown lining to cap (type I).			
	a. Finished appearance. The side crown lining shall be joined to the cap without twists, puckers, pleats, or distortions.			
	b. Position the left- and right-side crown lining, respectively, on the lining side of the top crown, with edges even and notches matching, and join to the top crown-side crown seam outlet with a 1/4-inch seam. The seam shall begin and end on the center fold of the top crown, 1/4 inch from each end.	301	Ssa-1	10-14
7	Join ends of side crown.			
	a. Finished appearance. The ends of the side crown shall be joined together without twists, puckers, pleats, or distortion. The center of the top crown shall be in line with the front and back seams of the cap. The stitching shall not extend to the top edge of the side crown.			
	b. Type I: Join the left- and right-side crowns at front and back ends of cap with a 3/8-inch seam at the lower end, tapering to 1/4 inch at the top crown joining stitch.	301	SSz-3(a)	10-14
	c. Type II: Join the left- and right-side crowns at front and back ends of cap with a 1/4-inch seam from the lower overedged edge up to the notch.	301	SSz-3(a)	10-14

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TABLE VI. Table of operations – Continued.

No.	Operation	Stitch Type	Seam Type	Stitches Per Inch
8	Join ends of side crown lining (type I).			
	a. Finished appearance. The ends of the side crown lining shall be joined without twists, puckers, pleats, or other distortions.			
	b. Join the left- and right-side crown lining at front and back ends of cap with a 3/8-inch seam at the lower end tapering to 1/4 inch at the top crown joining stitch.	301	Ssa-1	10-14
9	Topstitch side crown.			
	a. Finished appearance. The side crown shall be uniformly topstitched. The side crown lining at the front and back ends of the cap shall not be caught in the topstitching.			
	b. Turn the cap right side out, working out the upper front and back corners. Spread open the seam at each end of the cap and topstitch each side crown 1/16 inch from the seam. Turn the side-top crown joining seam toward the side and continue to topstitch each side crown 1/16 inch from the raised edge, catching the side-top crown joining seam in the stitching. NOTE: Type I: The topstitching at each end of the cap shall not be through the side crown lining.	301	SSz-3(b) ends LSq-2(b) raised edge	12-14
	c. On the finished cap, the width of the top crown shall be 3/4±1/8 inch, measured from raised edge of side crowns with the spread fully extended.			
10	Join lining to side crown (type I).			
	a. Finished appearance. The lining shall be joined to the side crown without twists, puckers, pleats, or other distortion. The front and back seam of lining and side crown shall match.			
	b. Turn cap wrong side out, matching the front and back lining seams to the respective crown seam. Join lining to side crown 1/4 inch from the raw bottom edge of the crown, with the front and back seams of side crown lining turned flat against the respective side crown seams and with the ends caught in the stitching.	301	Ssa-1	10 – 14
11	Join curtain to cap.			
	a. Finished appearance. The curtain shall fit close to the sides of the cap without pleats or puckers. The right and left ends of the curtain shall cross at the center from 1/2±1/8 inch below the top end of the side crown seam.			
	b. Turn the cap right side out and position the curtain on the outside with bottom edges even and notches in curtain matching the front seam.	101	Ssa-1	6-10

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TABLE VI. Table of operations – Continued.

No.	Operation	Stitch Type	Seam Type	Stitches Per Inch
11 (Cont'd)	c. Join the curtain to the cap, including through the lining for the type I cap, with a 3/8-inch seam, and with the front and back seams turned flat against the body of the cap and caught in the stitching.			
	d. On the finished cap, the left end of the curtain shall overlap the right end 4-1/2±1/4 inches, as indicated on figure 1, measured along the bottom finished edge of the cap.			
12	Join sweatband to cap. Note: Refer to operation 13 for alternate type II cap requirements.			
	a. Finished appearance. The sweatband shall be smoothly and evenly stitched to the cap, without wrinkles, puckers, gathers, twists, or other distortions. The sweatband shall cover the raw bottom edge of the curtain, with the lower edge of the sweatband 1/8 to 3/16 inch above the bottom finished edge of the cap. The ends of the sweatband shall finish to the right or left of, but not on, the back seam of the crown and shall lap each other not less than 3/4 inch with the top end turned under not less than 3/8 inch. The combination identification, instruction, and size label shall be placed to the right or left of the back seam of the crown with the top end under the sweatband.			
	b. The sweatband shall be 1±1/8 inch wide.	301	LSd-1 (each edge)	10-14
	c. Turn down the curtain and position the sweatband on the inside of the cap, with both side edges of sweatband turned under in line with the stitching joining the curtain and side crown. Stitch the sweatband through curtain and side crown 1/8 inch from each folded edge, catching the turned-under portion in the stitching and leaving one end free for attaching the label.	301	SSat-2	10-14
	d. Or, as an alternate to operation 12.c, the sweatband may be sewn to the cap with a double-needle machine, sewing both sides of the sweatband in one operation.	301	LSd-1	8-12
	e. Turn under and lap the free end of the sweatband on the opposite end. Place the top end of the label under the upper free edge of the sweatband. Secure the free end of the sweatband to the cap with a row of stitches 1/8 inch from the folded upper and lower edges, catching the turned-under portions and the label in the stitching.			
	f. Turn up the curtain.			
13	Join curtain to cap and attach sweatband and label (type II). For the type II cap, the following construction methods are allowed for joining the curtain to the cap and for attaching the sweatband and label:			

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TABLE VI. Table of operations – Continued.

No.	Operation	Stitch Type	Seam Type	Stitches Per Inch
13 (Cont'd)	a. Overedge stitch the side crown and curtain together at the bottom edges.	502 or 503	SSa-1	10-14
	b. Or, as an alternate to operations 11 and 12.a, overlap the inside of the side crown on the inside of the curtain 3/8, and stitch together 1/4 inch from the edge of the side crown, attaching the curtain to the side crown. On the finished cap, the left end of the curtain shall overlap the right end 4-1/2±1/4 inches, measured along the bottom finished edge of the cap.	TBD		
	c. Cut the sweatband to the required length and stitch ends together.	304	LSa-1	8-12
	(1) The ends of the artificial leather sweatband shall be abutted and joined with a row of zigzag stitching reinforced on the underside with a strip of the coated cloth.			
	(2) The ends of the interfaced cloth sweatband shall be overlapped no less than 3/4 inch and secured with stitching, with the outer edge folded under not less than 3/8 inch.	304		8-12
	d. The sweatband shall be zigzag stitched to the cap on the inside through the curtain seam allowances and side pieces, with the stitching on the bottom edge of the sweatband. The combination identification, instruction, and size label shall be caught behind the sweatband during this joining operation. The curtain shall be turned up. NOTE: The cap shall be completely set and pressed (see operation 15) before sewing the artificial leather sweatband.	301	SSa-1	8-12
	e. Turn down the curtain and stitch the curtain to the side crown 1/16 inch from the raised edge.			
14	Clean cap. a. Pull and remove all basting thread. Trim all ends of stitching and remove all loose ends of thread from the cap. Remove all spots, stains, shade tickets and marking without injury to the material.			
15	Press cap (see 3.7).			
	a. Finished appearance. The top crown shall not extend beyond the topstitched upper edge of the side crowns. The front and back of the cap curtain shall not be creased.			
	b. Insert a side shaper between the cap curtain and each side crown and spread the curtain smoothly and evenly on the shaper.			
	c. Press the cap smooth and flat with a heated pressing machine or heated pressing iron, carefully folding and creasing the bottom edge. The pressing shall be done in such a manner that the seams joining the side crowns to the top crown finish at the top edge of the cap and the sweatband is not be exposed beyond the bottom edge of the cap.			
	d. Remove all pressing marks.			

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TABLE VI. Table of operations – Continued.

No.	Operation	Stitch Type	Seam Type	Stitches Per Inch
16	Attach eyelet to curtain.			
	a. Finished appearance. The eyelet shall be properly positioned and securely attached through the outside of the left side of the curtain and securely clinched on the inside without distorting and cutting material or splitting the flange of the eyelet.			
	b. Mark the left side of the curtain 1-1/4±1/8 inches from the top edge and 2+1/4, -1/8 inches from the front of the cap when laid flat.			
	c. Attach the metal eyelet (and washer, if used) to the curtain at the mark.			

3.7 Pressing. All pressing operations specified in table VI shall be performed by using a heated pressing machine or heated pressing iron.

3.8 Finished dimensions. Finished dimensions of the finished cap shall be as shown in table VII when measured as specified in table VIII. All dimensions and tolerances are expressed in inches.

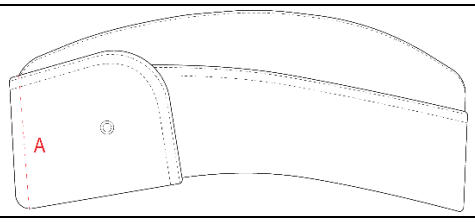
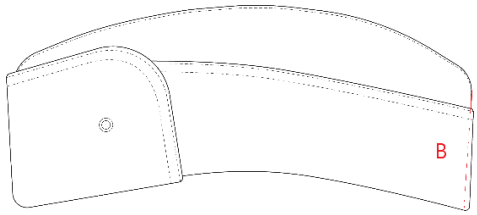
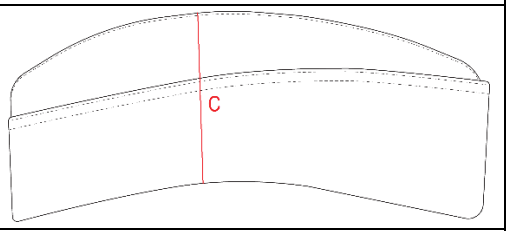
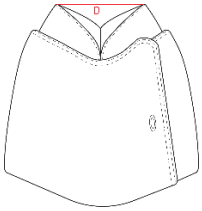
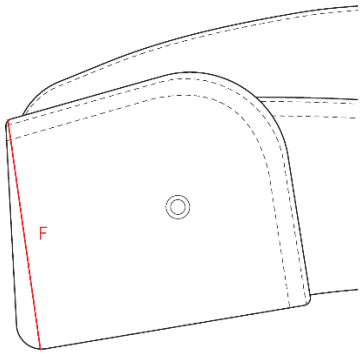
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TABLE VII. Table of finished dimensions.

Size	Height of Cap at Front (A)	Height of Cap at Back (B)	Height of Cap at Center (C)	Width of Top Crown (D)	Circumference (E)	Curtain Height Front (F)	Curtain Height Back (G)	Curtain Height Center, Left, and Right (H)	Through Eyelet (I)
21	2-7/8	2-7/8	3-3/8	4	21	2-3/4	2	2	2-3/4
21-1/2	2-7/8	2-7/8	3-3/8	4-1/8	21-1/2	2-3/4	2	2	2-3/4
22	3	3	3-1/2	4-1/4	22	2-7/8	2-1/8	2-1/8	2-7/8
22-1/2	3	3	3-1/2	4-3/8	22-1/2	2-7/8	2-1/8	2-1/8	2-7/8
23	3-1/8	3-1/8	3-1/2	4-1/2	23	3	2-1/4	2-1/4	3
23-1/2	3-1/8	3-1/8	3-5/8	4-5/8	23-1/2	3	2-1/4	2-1/4	3
24	3-1/8	3-1/8	3-5/8	4-3/4	24	3	2-1/4	2-1/4	3
Tolerance	±1/8	±1/8	±1/8	±1/8	±1/8	±1/8	±1/8	±1/8	±1/8

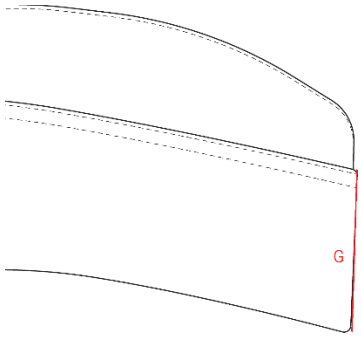
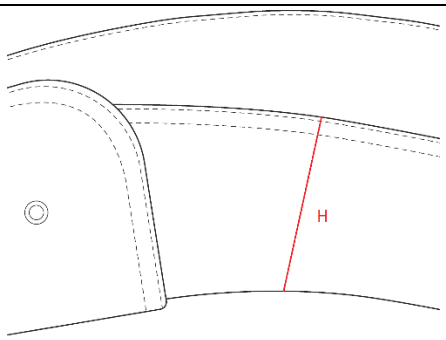
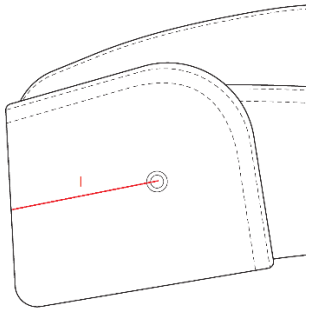
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TABLE VIII. Measurement descriptions.

Point of Measure	Measurement Description	Guidance
Height of Cap at Front (A)	Measurement of the height of the cap taken between the side crown and the curtain at the front seam, from bottom seam to top crown joining seam.	
Height of Cap at Back (B)	Measurement of the height of the cap taken between the side crown and the curtain at the back seam, from bottom seam to top crown joining seam.	
Height of Cap at Center (C)	Measurement of the height of the cap taken between the side crown and the curtain, halfway along the length of the cap [1/], from bottom seam to top crown joining seam.	
Width of Top Crown (D)	Measurement of the width of the top crown taken at the center of the crown from the raised edge to the raised edge of the side crown with the spread fully extended.	
Circumference (E)	Measurement of the circumference of the cap taken following the center of the sweatband. [2/]	As specified in measurement description
Curtain Height Front (F)	Measurement of the height of the curtain taken at the front seam.	

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TABLE VIII. Measurement descriptions – Continued.

Point of Measure	Measurement Description	Guidance
Curtain Height Back (G)	Measurement of the height of the curtain taken at the back seam.	
Curtain Height Center, Left, and Right (H)	Measurement of the height of the curtain taken at the side crown from the folded edge of the curtain to the upper edge of the curtain taken in a line perpendicular to the folded edge at the halfway point. 1/	
Through Eyelet (I)	Measurement of the distance from the front seam to the center of the eyelet taken in a line perpendicular to the front seam.	

FOOTNOTE:

1/ The halfway point is found when the cap is folded with the front seam and the back seam together.

2/ Any industry standard measurement method or tool may be used for the measurement of the circumference of the cap. Examples of industry standard tools include calibrated tape, ruler, and calipers, but other means may be used.

3.9 Toxicity. Unless otherwise specified in the contract or procurement documents (see 6.2), when tested as specified in 4.4.2, the finished cap shall not present a health hazard and shall show compatibility with prolonged, direct skin contact, as specified in 40 CFR § 798.2250, Dermal Toxicity, and 40 CFR § 798.4100, Dermal Sensitization. Chemicals recognized by the Environmental Protection Agency (EPA) as human carcinogens shall not be used.

3.10 Workmanship. The finished cap shall conform to the quality of product established by this document, be uniform, and be free from the defects specified in 4.5 and those defects that adversely affect form, fit, or function. Defects shall not exceed the established Acceptance Quality Limits (AQL) (see 6.2) and shall not adversely affect the serviceability, appearance, and uniformity of the product.

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on the cap when a first article sample is required (see 3.1.1). This inspection shall include the examinations and tests of 4.4 for compliance with the appearance, color, finish, design, configuration, workmanship, dimensional, and testing requirements as specified in this document.

4.2.1 First article samples. The first article sample size shall be as specified in the procurement document (see 6.2). The sample shall be one finished cap. The lot size shall be expressed in units of caps.

4.3 Conformance inspection. Conformance inspection shall include the examinations specified in 4.4.1 through 4.5.

4.3.1 Conformance inspection samples. Unless otherwise specified in the contract (see 6.2), sampling for inspection shall be performed in accordance with ASQ/ANSI Z1.4. The sample shall be one finished cap. The lot size shall be expressed in units of caps.

4.4 Inspection methods. Visual and dimensional examinations shall be made at any point or during any phase of the manufacturing process to determine whether construction details that cannot be examined in the finished product are in accordance with the requirements specified in section 3. Any in-process nonconformance remaining in the finished cap shall be classified as a defect.

4.4.1 Material and component examinations and tests. Unless otherwise excluded, amended, or modified in this document or applicable procurement documents (see 6.2), materials and components shall be examined and tested in accordance with the examinations and tests (see 6.7) specified in tables IX, X, and XI.

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TABLE IX. Component and material verification.

Material	Requirement	Verification
Basic material	3.3.2	MIL-DTL-10176 1/
Laminated basic material 2/	3.3.3	
Color		Visual
Colorfastness to drycleaning		AATCC TM132 3/ 4/ 5/
Dimensional stability		ASTM D2724 4/
Steam pressing (2 cycles)	3.3.3	Visual 4/ 5/
Bond strength, ounces per inch	3.3.3	ASTM D2724
Foam		
Weight, ounces per square yard	Table I	ASTM D3776/D3776M
Melt point, °C	Table I	AATCC TM20A
Knit		
Weight, ounces per square yard	Table I	ASTM D3776/D3776M
Pick count		
Wales	Table I	ASTM D3775
Courses	Table I	ASTM D3775
Colorfastness to perspiration (acid and alkaline)	Table I	AATCC TM15 3/ 5/ 6/
Knit type	Table I	Visual
Toxicity	3.9	4.4.2
Lining, polyester (alternate)	3.3.4.2	A-A-52106 1/
Lining, acetate (alternate)	3.3.4.3	Table X
Sweatband, cotton	3.3.5	MIL-DTL-10296 1/
Sweatband, cloth, interfaced 2/	3.3.5.1.2	MIL-DTL-10296 1/
Thread, covered polyester	3.3.6	A-A-50199 1/
Thread, shade	3.3.6	Visual
Thread, colorfastness to light	3.3.6.1	A-A-50199
Thread, colorfastness to drycleaning	3.3.6.1	A-A-50199
Eyelets	3.3.7	Table XI
Labels	3.3.8.1 3.3.8.2	MIL-DTL-32075 1/

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TABLE IX. Component and material verification – Continued.

FOOTNOTES:

1/ All applicable test methods cited in the document shall apply.

2/ Alternate for type II cap only.

3/ Rated using AATCC EP1.

4/ Tested after 3 drycleaning cycles.

5/ Samples shall be visually evaluated for degradation, bubbling, puckering, or delamination after every test or after each exposure to perspiration, cleaning, or pressing cycle.

6/ Steam pressing shall be conducted by wetting the cap with water and steam pressing at a temperature of 240 to 250 °F for 1 minute, then repeating one more time and visually checking for degradation after each exposure.

7/ Rated using AATCC EP2.

TABLE X. Acetate lining verification (see 3.3.4.3).

Characteristic	Verification
Fiber content	AATCC TM20A
Denier (minimum)	ASTM D1907/D1907M
Weave	Visual
Weight (minimum)	ASTM D3776/D3776M
Yarns per inch (minimum)	ASTM D3775
Warp	
Filling	
Breaking strength (minimum)	ASTM D5034
Warp	
Filling	
Dimensional stability (maximum)	AATCC TM158
Colorfastness to:	
Perspiration	AATCC TM15 1/
Drycleaning	AATCC TM132 2/
Crocking	AATCC TM8 3/
Wet (minimum)	
Dry (minimum)	

FOOTNOTES:

1/ Rated using AATCC EP1 and AATCC EP2.

2/ Rated using AATCC EP2 or AATCC TM8.

3/ Rated using AATCC EP8.

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TABLE XI. Eyelet and optional washer verification (see 3.3.5).

Characteristic	Verification
EYELET	
Material identification	Standard commercial
Flange type	Visual
Color (flange or entire eyelet)	Visual
Barrel	
Number of scores	Visual
Outside diameter	Standard commercial
Length (under flange)	Standard commercial
Flange	
Outside diameter	Standard commercial
Metal thickness	Standard commercial
WASHER (optional)	
Material identification	Standard commercial
Outside diameter	Standard commercial
Inside diameter	Standard commercial
Height (at highest point)	Standard commercial
Thickness	Standard commercial

4.4.1.1 Component and material sampling. Unless otherwise specified in referenced specifications, sampling shall be in accordance with table XII. The lot shall be unacceptable if one or more sample units fail to meet any test requirements specified.

TABLE XII. Material sampling.

Lot Size	Sample Size (Sample Unit)
800 or less	2
801 through 22,000	3
22,001 and over	5

The unit for expressing lot size and sample unit for testing each component shall be in accordance with the applicable referenced specification and as specified in table XIII.

TABLE XIII. Component sampling.

Component	Lot Size Expressed As	Sample Unit for Testing
Lining material	Yards	3
Artificial leather sweatband	Yards	2
Eyelet	Each	10

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4.4.2 Toxicity test. When the toxicity requirement (see 3.9) can be demonstrated with historical use data, on the finishing treatments used, toxicity testing may not be required. If dermal toxicity testing is required (see 6.2), it shall be conducted in accordance with 40 CFR § 798.2250 and 40 CFR § 798.4100, which are consistent with OECD Test No. 406 and OECD Test No. 404 (see 6.5).

4.5 End item examination.

4.5.1 Visual examination. The end item examination for the finished cap shall be as specified in MIL-STD-902 and shall include examination for the barcode label/tag defects listed below. AQLs shall be as specified in the contract or order (see 6.2).

a. Barcode omitted or not readable by scanner – minor

b. Human Readable Interpretation (HRI) omitted or illegible – minor

c. Not visible on folded, packaged item – minor

d. Causes damage to the item – major

4.5.2 Dimensional examination. The cap shall be examined for conformance with the dimensions specified in table VII. Measurement locations are indicated in 3.8 and table VII. Any measurement deviating from dimensions and tolerances specified in table VII shall be scored as a measurement defect.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The cap is intended for wear by female personnel of the U.S. Marine Corps.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type, class, and size required (see 1.2).

c. NSNs.

d. When a first article sample and inspection is required and first article sample requirements (see 3.1.1 and 4.2.1).

e. Requirements for conformance inspection and conformance inspection sampling (see 3.1.2 and 4.3.1).

f. Applicable Government patterns (see 3.4).

g. Toxicity testing requirements (see 3.9 and 4.4.2).

h. Acceptance quality limits (AQLs) (see 3.10 and 4.5.1).

i. Any deviations from the inspection and test procedures in the referenced documents (see 4.4.1).

j. Packaging requirements (see 5.1).

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6.3 Information requests. Requests for information such as purchase descriptions, patterns, drawings, standard shade samples of cloth or buttons can be made by completing and submitting the Defense Logistics Agency Troop Support's Clothing and Textile Specification/Pattern/Drawing Request form online at https://www.dla.mil/TroopSupport/ClothingandTextiles/SpecRequest. Requests to use equivalent materials or components or to make changes to the pattern should be sent to the contracting officer for approval by the military services.

6.4 Gages. Measuring gages are available online through several sources, and any reliable measuring gage may be used. Information concerning the measuring gage may be obtained upon request to the contracting officer.

6.5 Superseding data for class designations. In the original document, the classes were designated by basic material. Class 1 caps were lined, and class 2 caps were unlined. In this revision, the wool basic materials previously cited for class 2 have been deleted. As specified in table XIV, all caps are now class 1, and type I is the lined cap and type II is an unlined cap for the optional (certified item) program.

TABLE XIV. Classification designations.

Type/Class	MIL-C-29431	MIL-DTL-29431A
Type I	Applicable to all caps	Type added to designate lined cap
Type II	N/A	Type added to designate unlined cap (optional program)
Class 1	Cap, Polyester/Wool, Gabardine, Green 2212, lined	Cap, Polyester/Wool, Gabardine, Green shade 2212
Class 2	Cap, Gabardine, Wool, Green 2210, unlined	Deleted

6.6 Toxicity testing. OECD Test No. 406 and OECD Test No. 404 have been suggested to assess this requirement, but other methods may be used.

6.7 Artificial leather sweatband. Before an artificial leather sweatband is used, a sample of the material should be provided to the contracting agency for approval, as several materials have exhibited durability issues in cleaning.

6.8 Certificates of compliance (CoC). CoC may be accepted by the contracting agency in lieu of specified testing on end items when testing of the end item is difficult. Vendors should confirm acceptability of CoC prior to submitting them in lieu of testing.

6.9 Subject term (key word) listing.

Acetate
Curtain
Gabardine
Laminate
Unlined

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6.10 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

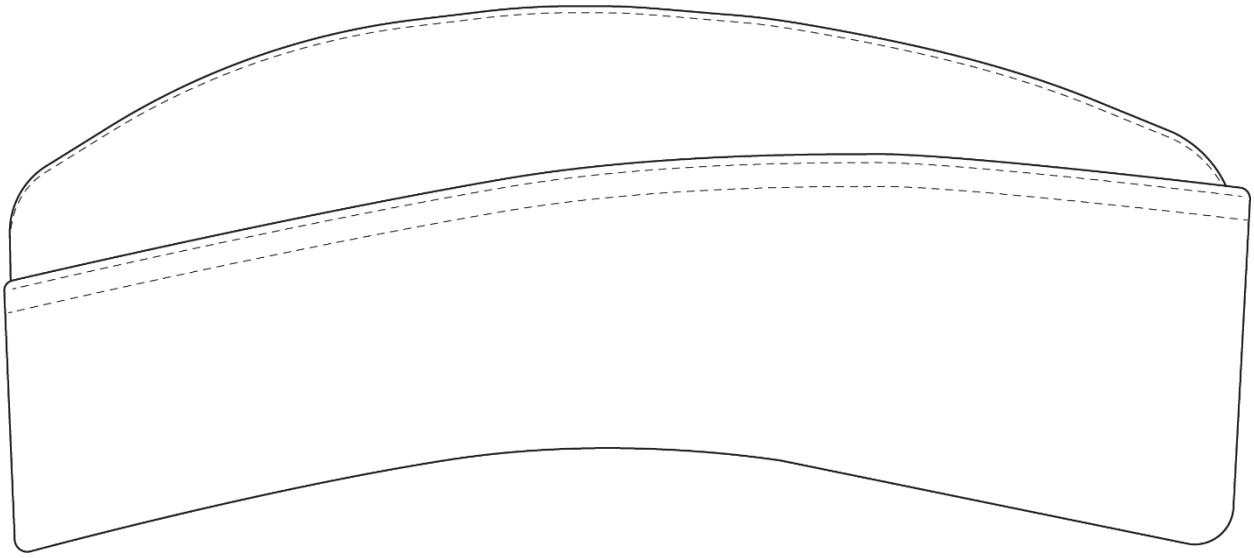


FIGURE 1. Women's garrison cap – right side.

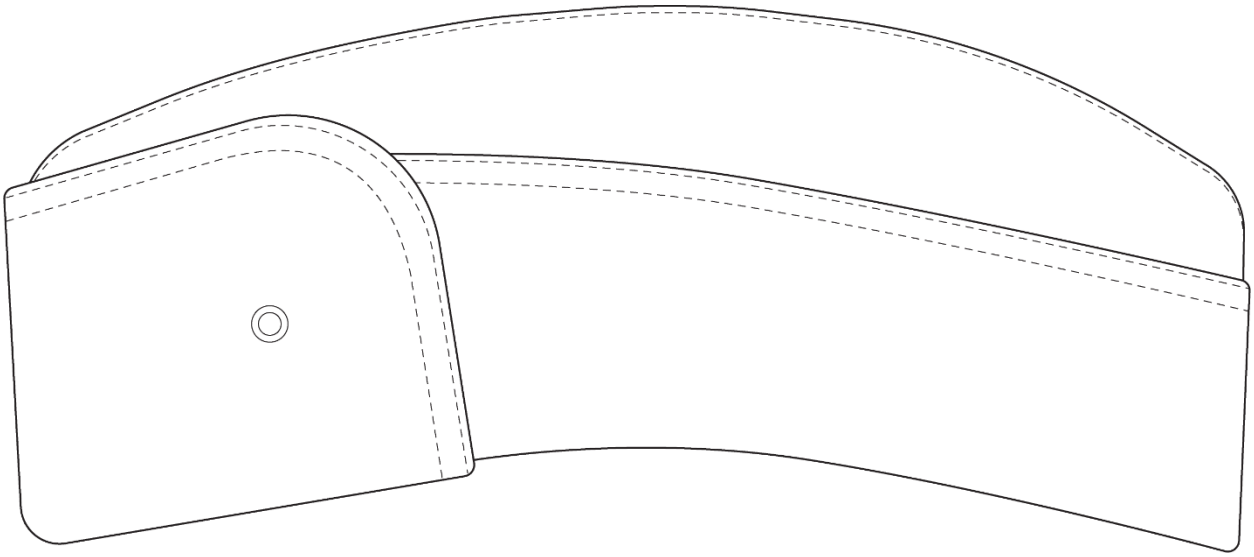


FIGURE 2. Women's garrison cap – left side.

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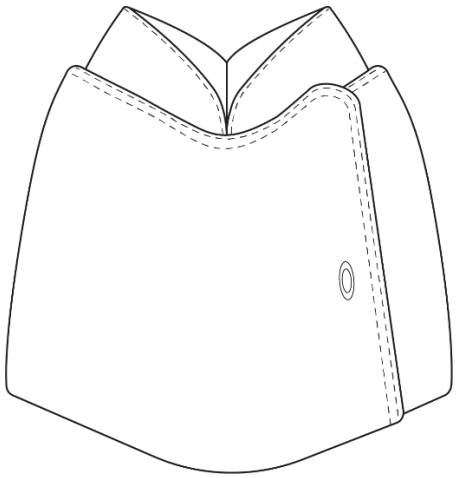


FIGURE 3. Women's garrison cap, class 1 – front.

CONCLUDING MATERIAL

Custodian:
Navy – MC

Preparing activity:
Navy – MC
(Project 8410-2023-003)

Review activity:
Navy – NU

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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